

Phishing Website Detection Using Machine Learning Models

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Abstract—Phishing websites remain one of the most common vectors used to steal credentials and financial information from online users. This project evaluates the effectiveness of six supervised machine learning models – Logistic Regression, Softmax (Multinomial) Regression, K-Nearest Neighbors, Decision Tree, Random Forest, and a Multilayer Perceptron Neural Network – in classifying websites as phishing or legitimate using the UCI Phishing Websites dataset. Principal Component Analysis (PCA) was additionally applied to examine feature redundancy and dimensionality reduction trade-offs. Random Forest achieved the highest accuracy (94.19%) and AUC (0.988), while Logistic Regression and Softmax Regression scored lowest (91.28%). PCA showed that 23 of the 30 original features are needed to retain 95% of the variance, and reduced-dimensionality features led to a small drop in Random Forest accuracy (94.19% to 92.39%), indicating limited feature redundancy in this dataset.

Index Terms—phishing detection, machine learning, logistic regression, random forest, decision tree, KNN, neural network, softmax regression, principal component analysis

I. INTRODUCTION

A. Problem Statement

Phishing is a form of social engineering attack in which a malicious actor impersonates a legitimate organization, typically through a counterfeit website, to trick users into revealing sensitive information such as login credentials, banking details, or personal identification data. Because phishing websites are designed to visually and structurally mimic legitimate ones, manual identification is unreliable, and traditional blacklist-based filtering fails to catch newly created (“zero-hour”) phishing pages. This project addresses the problem of automatically distinguishing phishing websites from legitimate ones using structural and lexical features extracted from a website’s URL and metadata, framed as a supervised binary classification task.

B. Project Objectives

The key goals of this project are to:

- Perform exploratory data analysis and preprocessing on a labeled phishing website dataset, including handling duplicates and class imbalance checks.
- Apply Principal Component Analysis (PCA) to assess feature redundancy and the impact of dimensionality reduction on classification performance.

- Train and evaluate six machine learning classifiers – Logistic Regression, Softmax Regression, K-Nearest Neighbors, Decision Tree, Random Forest, and a Multilayer Perceptron – on the phishing detection task.
- Compare the models using accuracy, precision, recall, F1 score, confusion matrices, and ROC-AUC to identify the best-performing approach.

C. Literature Review

Reviewed by Tahsin Tausif:

Mohammad et al. [1] introduced the dataset used in this project by proposing an automated technique to identify and validate 30 URL, domain, and page-based features that distinguish phishing websites from legitimate ones, using rule-induction (PRISM) to confirm feature relevance rather than optimizing for raw accuracy. Their labeled 11,055-instance dataset became a widely reused benchmark in subsequent phishing detection research, including this project. A limitation is that the underlying features are largely static and lexical/host-based, meaning they can become outdated as browser security indicators, certificate practices, and hosting patterns evolve. The paper also leaves systematic comparison of modern classifiers and ensembles as an open gap for later work to fill.

Sahingoz et al. [2] proposed a real-time phishing detection system using only URL-based lexical, word-vector (NLP), and host-based features, purposely excluding third-party services (e.g. search-engine ranking or WHOIS lookups) so predictions can be made instantly. Seven classifiers were compared, with Random Forest reaching the highest accuracy of 97.98%. A limitation is that purely URL-based features cannot capture page-content-level phishing cues, and the NLP feature extraction adds computational overhead compared to simpler lexical features. The paper does not explore deep learning approaches, leaving a gap that later CNN/embedding-based studies address.

Chiew et al. [3] proposed a hybrid ensemble feature selection framework (HEFS) that combines multiple filter-based feature-ranking methods with a Data Perturbation Ensemble to select a compact, low-redundancy feature subset before classification. Using Random Forest on the selected features, they achieved 94.6% accuracy while using only a fraction

of the original feature set, demonstrating that careful feature selection can match or exceed full-feature-set performance. A limitation is that the ensemble feature-selection process is computationally more expensive than using a fixed feature set, and the approach was validated primarily on one dataset, leaving generalizability across different phishing feature sets as a research gap.

Rao and Pais [4] developed Jail-Phish, a search-engine-based phishing detection system that improves on prior search-engine approaches by also correctly identifying phishing pages hosted on compromised legitimate domains, which earlier systems misclassified. The system combines search-engine result matching with hyperlink- and target-domain-based features, reporting accuracy above 98% on their evaluation set. A key limitation is its dependency on live search engine queries and network access at prediction time, making it slower and less suitable for real-time or offline detection. The paper also notes reduced reliability for very newly registered domains with little to no search engine presence, a gap this project avoids by relying only on precomputed static features.

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Jain and Gupta [5] proposed a phishing detection approach based solely on hyperlink-based features extracted from a webpage's HTML source, such as the ratio of internal to external links, empty links, and links in the `<meta>`, `<script>`, and `<link>` tags. Using Random Forest as the primary classifier, they achieved 98.4% accuracy without relying on third-party services such as search engines or WHOIS records, making the method fast and self-contained. A limitation is that the approach requires the full HTML source to be fetched before classification, which is not always feasible for real-time filtering, and it can be evaded by attackers who mimic legitimate hyperlink structures, a gap later addressed by combining hyperlink features with URL-lexical features.

Yerima and Alzaylaee [6] applied a Convolutional Neural Network (CNN) directly to structured phishing website features rather than using traditional feature-based classifiers, framing feature vectors as input "images" for the CNN. Their model achieved a phishing detection rate of 98.2% and an F1-score of 0.976, outperforming traditional machine learning classifiers evaluated on the same dataset. A limitation of this deep learning approach is reduced interpretability compared to tree-based models, and the higher computational cost of training CNNs compared to classical classifiers. The paper does not explore model explainability, which remains a gap for deployment in security-sensitive settings requiring justified decisions.

Taha [7] proposed an ensemble learning approach for phishing detection based on weighted soft voting, where four heterogeneous base classifiers are combined using weights derived from each classifier's Kappa statistic, giving more influence to stronger learners rather than treating all base models equally. This weighted ensemble outperformed both the individual base classifiers and standard majority voting in the reported experiments. A limitation is the added complexity of tuning and computing per-classifier weights compared to

simple ensembles, and the approach was evaluated on a single dataset. The paper leaves evaluation on multiple, more diverse phishing datasets as an open direction.

Tang and Mahmoud [8] presented a broad survey of machine learning-based phishing website detection methods, covering the phishing lifecycle, common anti-phishing strategies, and a structured comparison of data collection, feature extraction, modeling, and evaluation approaches used across the literature. The survey highlights that Random Forest and other ensemble methods consistently perform well across studies, while deep learning methods show promise but at higher computational cost. A limitation inherent to survey papers is the lack of new experimental results; performance figures are only as reliable as the studies they summarize. The authors note a persistent research gap in evaluating models against continuously evolving, real-time phishing campaigns rather than static datasets.

D. Dataset Identified

Name: Phishing Websites Dataset (UCI Machine Learning Repository, Dataset ID 327).

Source: <https://archive.ics.uci.edu/dataset/327/phishing>

Format: Tabular data distributed as .arff/CSV, suitable for direct use with standard machine learning libraries.

Description: The dataset contains 11,055 website instances collected from the PhishTank archive, MillerSmiles archive, and Google's search operators, introduced by Mohammad et al. [1]. Each instance has 30 integer-valued features derived from the website's URL, domain, and page structure (e.g. `having_IP_Address`, `URL_Length`, `SSLfinal_State`, `age_of_domain`, `web_traffic`), grouped conceptually into address-bar-based, abnormal-based, HTML/JavaScript-based, and domain-based feature categories. The target variable, `Result`, labels each instance as phishing (-1) or legitimate (1); for this project the labels were remapped to 0 and 1 respectively. After removing 3,206 duplicate rows, 5,849 unique instances remained (3,019 legitimate, 2,830 phishing), which were used for model training and evaluation.

II. METHODOLOGY

A. Dataset Description

As detailed in Section I-D, the dataset used is the UCI Phishing Websites dataset, comprising 30 features extracted from a website's URL and page structure, with a binary target indicating phishing or legitimate status. Fig. 1 shows the pairwise correlation matrix of all numerical features after preprocessing; most feature pairs show weak correlation, with a small number of moderate positive clusters (e.g. among URL and anchor-based features), suggesting limited multicollinearity overall – consistent with the PCA finding in Section II-C that 23 of the 30 components are needed to retain 95% of the variance.

B. Data Preprocessing

The dataset was first checked for missing values, of which there were none. A total of 3,206 exact duplicate rows were

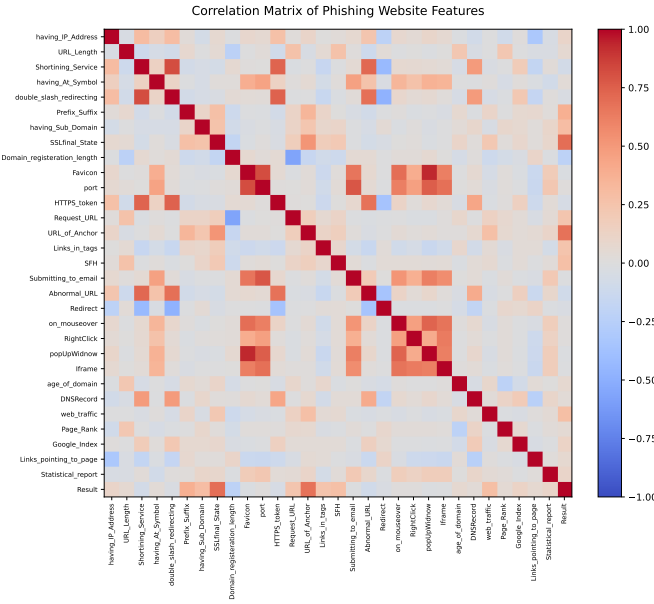


Fig. 1. Correlation matrix of the phishing website dataset features.

identified and removed, reducing the dataset from 11,055 to 5,849 instances. The target column was remapped from $\{-1, 1\}$ to $\{0, 1\}$ for compatibility with scikit-learn’s binary classifiers. The class distribution after cleaning was reasonably balanced (51.6% legitimate, 48.4% phishing), so no resampling technique (e.g. SMOTE) was required. The cleaned dataset was split into training and testing sets using an 80–20 stratified split (`random_state=42`), yielding 4,679 training and 1,170 testing instances.

C. Dimensionality Reduction: Principal Component Analysis

PCA was applied to the standardized feature set to evaluate feature redundancy. Fitting PCA with all 30 components showed that 23 components were required to retain 95% of the total variance, indicating that the feature set has only modest redundancy. A 2D projection onto the first two principal components showed partial, but incomplete, visual separation between phishing and legitimate classes. To quantify the practical effect of dimensionality reduction, a Random Forest classifier was trained on both the full 30-feature set and the 23-component PCA-reduced representation; accuracy dropped from 94.19% to 92.39%, confirming that a meaningful portion of predictive information is spread across the lower-variance components rather than concentrated in a small number of them.

D. Learning Phase

Six classifiers were trained on the preprocessed dataset and evaluated on the held-out test set.

a) Logistic Regression: A linear model that estimates the probability of a website being phishing via the sigmoid function applied to a weighted sum of input features. It was trained directly on the unscaled feature set.

b) Softmax (Multinomial) Regression: A generalization of logistic regression that models class probabilities using the softmax function; for this binary target it converges to the same decision boundary as standard logistic regression, but the formulation generalizes to multi-class problems. It was trained on standardized features using the `lbfgs` solver.

c) K-Nearest Neighbors (KNN): A non-parametric, distance-based classifier that assigns a class label based on the majority vote of the $k = 5$ nearest training instances in feature space. Because KNN relies on distance calculations, features were standardized beforehand to prevent large-scale features from dominating the distance metric.

d) Decision Tree: A tree-based classifier that recursively partitions the feature space by selecting the most informative splitting attribute at each node. A maximum depth of 10 was used to limit overfitting.

e) Random Forest: An ensemble of 100 decision trees, each trained on a bootstrapped sample of the data with random feature subsampling at each split, with final predictions aggregated by majority vote. This reduces the variance and overfitting risk associated with individual decision trees.

f) Multilayer Perceptron (MLP): A feedforward neural network with two hidden layers (32 and 16 neurons) trained via backpropagation on standardized features, capable of learning non-linear decision boundaries that the linear models cannot represent.

III. RESULTS AND DISCUSSION

A. Dimensionality Reduction Results

The PCA scree plot showed a gradual, near-linear increase in cumulative explained variance rather than a sharp elbow, consistent with the finding that 23 of 30 components were needed for 95% variance retention. This suggests that, unlike datasets with strong inter-feature correlation, the phishing feature set contributes comparatively independent information across most of its 30 features, which is consistent with the feature set being deliberately curated from several conceptually distinct categories (address-bar, abnormal, HTML/JavaScript, and domain-based features) by the original dataset authors [1].

B. Performance Metrics

Table I summarizes the accuracy, precision, recall, F1 score, and AUC for all six models on the test set (1,170 instances).

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Acc.	Prec.	Rec.	F1	AUC
Logistic Regression	0.9128	0.9000	0.9223	0.9110	0.9738
Softmax Regression	0.9128	0.9000	0.9223	0.9110	0.9738
KNN	0.9188	0.9213	0.9099	0.9156	0.9724
Decision Tree	0.9342	0.9454	0.9170	0.9309	0.9687
Neural Network (MLP)	0.9265	0.9167	0.9329	0.9247	0.9863
Random Forest	0.9419	0.9368	0.9435	0.9401	0.9881

Random Forest achieved the highest scores across accuracy, F1, and AUC, consistent with the literature reviewed in Section

I-C, where ensemble tree-based methods repeatedly outperformed single linear or distance-based classifiers on similar phishing feature sets [2], [3], [8]. Logistic Regression and Softmax Regression produced identical results, which is expected: with a binary target, the softmax formulation mathematically reduces to the sigmoid-based logistic regression decision boundary, so no performance difference should be expected between the two, and the observed match confirms a correct implementation rather than a labeling error. The Decision Tree achieved the highest precision (0.9454) but a comparatively lower recall (0.9170), meaning it is more conservative in labeling a site as phishing than Random Forest, which trades a small amount of precision for meaningfully higher recall (0.9435) – an important consideration given that a missed phishing site (false negative) is generally more costly to a user than a legitimate site being flagged for review (false positive). The confusion matrices in Fig. 2 and the ROC curves obtained during evaluation (see accompanying notebook) support this: the Decision Tree’s higher false-negative count (phishing sites misclassified as legitimate) relative to Random Forest visually confirms its lower recall, while Random Forest’s ROC curve dominates the other five across nearly the entire false-positive-rate range, matching its highest AUC of 0.9881.

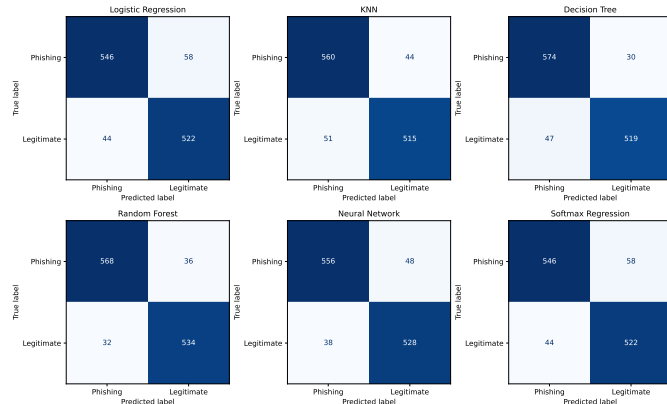


Fig. 2. Confusion matrices for all six models on the test set (0 = Phishing, 1 = Legitimate).

IV. CONCLUSION

This project evaluated six machine learning models – Logistic Regression, Softmax Regression, KNN, Decision Tree, Random Forest, and a Multilayer Perceptron – on the task of phishing website detection using the UCI Phishing Websites dataset. Random Forest was consistently the strongest performer across accuracy, F1 score, and AUC, aligning with prior literature on ensemble methods for this task. Principal Component Analysis showed that the dataset’s 30 features carry comparatively independent information, with 23 components required to retain 95% of the variance, and dimensionality reduction came at a measurable accuracy cost rather than being free redundancy removal. Future work could explore hybrid and stacked ensemble methods as demonstrated in [3], [7], hyperparameter tuning via grid or random search, and

evaluation on more recent phishing datasets to test whether these results generalize beyond the relatively dated feature set used here.

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